

FARM MANAGEMENT SYSTEM

PROJECT DOCUMENTATION

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TEAM MEMBERS: R . Pazhanivel , M . periyasamy , A . Nagaraj , K . pavithara

1. ABSTRACT

In the Farm Management System (FMS), as a farmer, farm manager, or system administrator, I want to use a centralized Salesforce-based platform to record and manage farm, crop, farmer, and buyer information so that all agricultural data is stored securely and systematically. I want to enter and update farm details, crop production records, and buyer transactions easily through a user-friendly interface with proper validation rules to prevent incorrect entries. I also want the system to generate real-time reports and interactive dashboards so that I can analyze farm productivity, monitor sales, and make informed decisions. Additionally, I want role-based access and secure data storage to ensure that sensitive information is protected while enabling smooth collaboration among users. This system will help reduce manual paperwork, minimize errors, improve efficiency, and support data-driven farm management.

2. INTRODUCTION

Farm Management System (FMS) is a Salesforce-based application designed to digitize and streamline agricultural

operations. It manages data related to farms, crops, farmers, and buyers while providing real-time insights through reports and dashboards. The system helps farmers and managers monitor agricultural activities efficiently and make informed decisions

3. Purpose

The main purpose of this project is to:

- **Digitize agricultural data management**
- **Improve data accuracy and accessibility**
- **Automate record handling**
- **Enable real-time monitoring**
- **Support data-driven farming decisions**
- **Enhance productivity and transparency**

4. Data Flow Diagram

Data Flow:

User → Salesforce UI → Custom Objects (Farm, Crop, Farmer, Buyer) →

Validation Rules → Relationships → Reports → Dashboards

5. PROPOSED SYSTEM

The Farm Management System (FMS) in Salesforce is a comprehensive solution designed to digitize and streamline agricultural operations. This application empowers farmers, farm managers, and stakeholders by managing data related to crops, Farm, Farmer, and buyers, while providing actionable insights through reports, and dashboards.

6. Project Planning

- **Created Salesforce Developer Org**
- **Developed custom objects**
- **Configured tabs and Lightning App**
- **Created fields and relationships**
- **Designed profiles and roles**
- **Created users and sharing rules**
- **Developed reports and dashboards**
- **Conducted testing and reviews**

7. FUNCTIONAL & PERFORMANCE TESTING

During the testing phase, screenshots were captured from the Salesforce interface to verify system functionality. The following were tested:

- **Creation of Farm, Crop, Farmer, and Buyer records**
- **Validation of input fields**
- **Relationship and dependency checks**
- **Profile and role access testing**
- **Sharing rule verification**
- **Report and dashboard generation**

Performance testing ensured smooth system response and stability during multiple user operations.

8. ADVANTAGES & DISADVANTAGES

Advantages

- **Centralized data system**
- **Reduced paperwork**
- **Improved accuracy**
- **Better monitoring**
- **Real-time analytics**

Disadvantages

- **Requires internet connection**
- **Salesforce license dependency**
- **Limited offline access**

9. HARDWARE REQUIREMENTS

Processor: Intel Core i3 or above

RAM: 4 GB or above

Hard Disk: 250 GB or above

Keyboard and Mouse

Monitor

10 .Conclusion:

The Farm Management System successfully digitizes agricultural data management using Salesforce. It improves efficiency, reduces errors, and enables farmers and managers to make informed decisions through automation and analytics.

11. FUTURE SCOPE

- **Mobile application integration**
- **AI-based crop prediction**
- **Weather data integration**
- **IoT sensor connectivity**
- **Government scheme integration**
- **Market price forecasting**

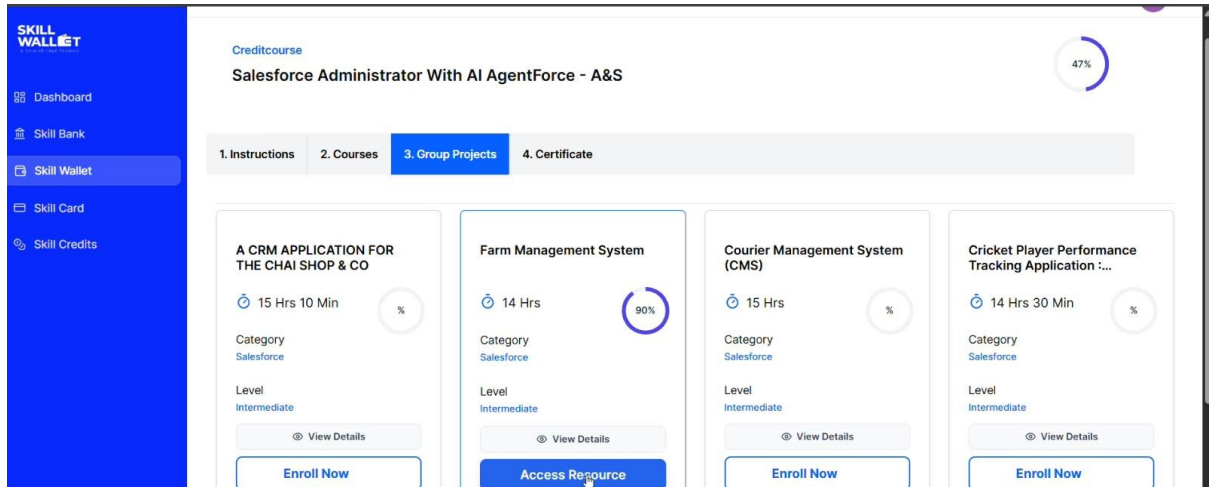
12. APPENDIX

- **Source Code: Salesforce Configuration and Metadata**
- **Dataset: Farm, Crop, Farmer, and Buyer Records**
- **Tools Used: Salesforce Developer Org**

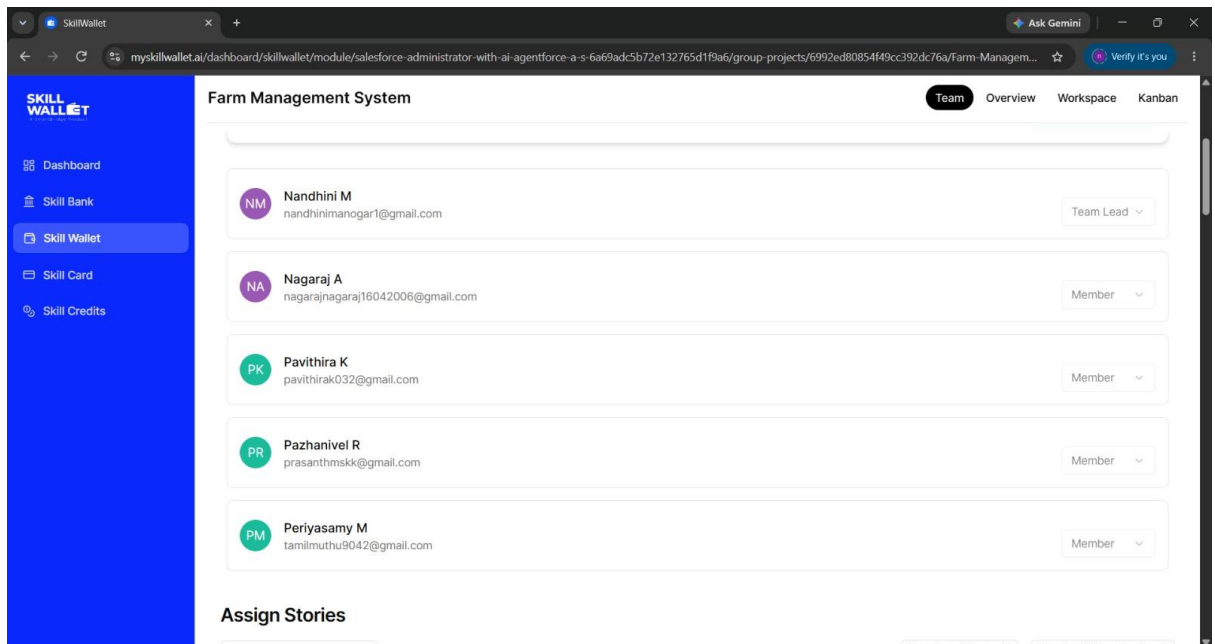
Thank You

14. PROJECT SCREENSHOTS

Screenshot 1 – Project Dashboard



Screenshot 2 – Team Members



Screenshot 3 – Project Overview

This screenshot shows the 'Project Overview' page for the 'Farm Management System' in the SkillWallet application. The page is divided into several sections:

- Header:** The top navigation bar includes 'Team', 'Overview' (selected), 'Workspace', and 'Kanban'.
- Left Sidebar:** Contains navigation links for 'Dashboard', 'Skill Bank', 'Skill Wallet' (selected), 'Skill Card', and 'Skill Credits'.
- Main Content Area:**
 - Project Description:** A paragraph explaining the Farm Management System (FMS) in Salesforce, designed to digitize and streamline agricultural operations.
 - Skills Required:** A list of skills including 'Salesforce', 'Data Modelling', 'Data Security', 'Data Validation', and 'Salesforce Lightning'.
 - Mentor:** A section stating 'No mentor assigned yet'.
 - Team Members:** A list of team members: Nandhini M (Team Lead), Pavithira K (Member), Periyasamy M (Member), Nagaraj A (Member), and Pazhanivel R (Member).
 - Project Stats:** A section for project statistics.
- Right Sidebar:**
 - Project Progress:** A progress bar showing 90% completion, with buttons for 'View Demo' and 'View Repo'.
 - User Story:** A section for user stories, including 'Salesforce Account Creation' and 'Objects'.

Screenshot 4 – Workspace and Project Progress

This screenshot shows the 'Workspace' view of the 'Farm Management System' project in SkillWallet. The page is divided into several sections:

- Header:** The top navigation bar includes 'Team', 'Overview', 'Workspace' (selected), and 'Kanban'.
- Left Sidebar:** Contains navigation links for 'Dashboard', 'Skill Bank', 'Skill Wallet' (selected), 'Skill Card', and 'Skill Credits'.
- Main Content Area:**
 - User Story:** A section for user stories, including 'Salesforce Account Creation' and 'Objects'.
 - Description:** A detailed description of the Farm Management System (FMS) project, explaining its purpose and goals.
 - Resources:** A section for resources, currently showing 'No resources available'.
- Right Sidebar:**
 - Project Progress:** A progress bar showing 90% completion, with buttons for 'View Demo' and 'View Repo'.
 - User Story:** A section for user stories, including 'Salesforce Account Creation' and 'Objects'.